

Project Documentation: GOG Games — Rating Prediction & EDA

Dataset: <https://www.kaggle.com/datasets/lunthu/gog-com-video-games-dataset/data>

About the data:

No. of columns in Raw Data: 51

No. of rows in Raw Data: 10896

Why is Review prediction important?

Game review prediction is critical because it directly influences a game's market success and shapes ongoing development. By forecasting user ratings before or immediately upon launch, developers can optimize marketing strategies, address critical player feedback, and refine gameplay to ensure long-term engagement and profitability.

In this dataset, the target column is “overallAvgRating” (a continuous score from 0–5). For this project, the rating was bucketed into three quartile classes (rating_class ;Poor,Average And Excellent) and predicted as a multi-class classification problem.

1. Import the necessary libraries:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
pd.set_option('display.max_columns', 500)
```

2. Import Data: `df = pd.read_csv(“filepath”)`
Import Data: `df_orignal = pd.read_csv(“filepath”)` # To use later in EDA
3. Get a feel for the data using `df.head()` and `df.tail()`
4. Find shape and size of the data(use `df.shape` or `df.size`)

5. Use Info to see the name of each column the total non-null values

```
1 <class 'pandas.DataFrame'>
2 RangeIndex: 10896 entries, 0 to 10895
3 Data columns (total 51 columns):
4 #   Column                                Non-Null Count  Dtype
5 ---  ---                                -
6 0   id                                    10896 non-null  int64
7 1   developer                            10896 non-null  str
8 2   publisher                            10896 non-null  str
9 3   gallery                              10896 non-null  str
10 4   video                                6961 non-null   str
11 5   supportedOperatingSystems            10896 non-null  str
12 6   genres                               10896 non-null  str
13 7   globalReleaseDate                   10393 non-null  float64
14 8   isTBA                                10896 non-null  bool
15 9   isDiscounted_main                    10896 non-null  bool
16 10  isInDevelopment                      10896 non-null  bool
17 11  releaseDate                          10393 non-null  float64
18 12  availability                          10896 non-null  str
19 13  salesVisibility                      10896 non-null  str
20 14  buyable                              10896 non-null  bool
21 15  title                                10896 non-null  str
22 16  image                                10896 non-null  str
23 17  url                                  10896 non-null  str
24 18  supportUrl                           10896 non-null  str
25 19  forumUrl                             10896 non-null  str
26 20  worksOn                              10896 non-null  str
27 21  category                             10896 non-null  str
28 22  originalCategory                     10896 non-null  str
29 23  rating                               10896 non-null  int64
30 24  type                                 10896 non-null  int64
31 25  isComingSoon                         10896 non-null  bool
32 26  isPriceVisible                       10896 non-null  bool
33 27  isMovie                              10896 non-null  bool
34 28  isGame                               10896 non-null  bool
35 29  slug                                  10896 non-null  str
36 30  isWishlistable                       10896 non-null  bool
37 31  ageLimit                             10896 non-null  int64
38 32  boxImage                             10896 non-null  str
39 33  isMod                                10896 non-null  bool
40 34  currency                             10896 non-null  str
41 35  amount                               10896 non-null  float64
42 36  baseAmount                           10896 non-null  float64
43 37  finalAmount                          10896 non-null  float64
44 38  isDiscounted                         10896 non-null  bool
45 39  discountPercentage                   10896 non-null  int64
46 40  discountDifference                   10896 non-null  float64
47 41  isFree                               10896 non-null  bool
48 42  discount                             10896 non-null  int64
49 43  promoId                              7908 non-null   str
50 44  filteredAvgRating                    10896 non-null  float64
51 45  overallAvgRating                     10896 non-null  float64
52 46  reviewCount                          10896 non-null  int64
53 47  isReviewable                         10896 non-null  bool
54 48  reviewPages                          10896 non-null  int64
55 49  dateGlobal                           10393 non-null  str
56 50  dateReleaseDate                      10393 non-null  str
57 dtypes: bool(13), float64(8), int64(8), str(22)
58 memory usage: 21.1 MB
59
```

Observations:

- genres, worksOn, availability, salesVisibility are stored as strings that look like lists/dicts — ISSUE
- developer and publisher are raw text with hundreds of unique values — ISSUE
- globalReleaseDate is a Unix timestamp, not a usable year — ISSUE
- Several columns are pure metadata (image/gallery/video URLs, forum/support links, slugs) — not useful for prediction and were dropped
- Several NULL values — ISSUE

Checking for null and duplicated Values:

Checking for null values using
df.isnull().sum():

```
1  id 0
2  developer 0
3  publisher 0
4  gallery 0
5  video 3935
6  supportedOperatingSystems 0
7  genres 0
8  globalReleaseDate 503
9  isTBA 0
10 isDiscounted_main 0
11 isInDevelopment 0
12 releaseDate 503
13 availability 0
14 salesVisibility 0
15 buyable 0
16 title 0
17 image 0
18 url 0
19 supportUrl 0
20 forumUrl 0
21 worksOn 0
22 category 0
23 originalCategory 0
24 rating 0
25 type 0
26 isComingSoon 0
27 isPriceVisible 0
28 isMovie 0
29 isGame 0
30 slug 0
31 isWishlistable 0
32 ageLimit 0
33 boxImage 0
34 isMod 0
35 currency 0
36 amount 0
37 baseAmount 0
38 finalAmount 0
39 isDiscounted 0
40 discountPercentage 0
41 discountDifference 0
42 isFree 0
43 discount 0
44 promoId 2988
45 filteredAvgRating 0
46 overallAvgRating 0
47 reviewCount 0
48 isReviewable 0
49 reviewPages 0
50 dateGlobal 503
51 dateReleaseDate 503
52 dtype: int64
```

Checking for duplicate rows using `df[df.duplicated(keep=False)]`:

O/P: no fully duplicated rows were found.

Handling nulls:

- Remaining nulls (mostly in the timestamp columns and video, which were being dropped anyway): handled by a single `df.dropna()` after the irrelevant columns were removed.

Dropping columns that are irrelevant:

- Columns that contain metadata, duplicated columns were removed since they don't have any significance.
- RUN:

```
• df.drop(columns=[
    'id', 'video', 'gallery', 'image', 'dateReleaseDate', 'releaseDate',
    'dateGlobal', 'boxImage', 'url', 'title', 'slug', 'supportUrl', 'forumUrl',
    'supportedOperatingSystems', 'originalCategory', 'baseAmount', 'amount',
    'currency', 'rating', 'filteredAvgRating', 'discountPercentage',
    'isDiscounted', 'isDiscounted_main', 'promoID'
], inplace=True)
```

6. Data cleaning

a. Genres — multi-label binarization

- genres is a list of tags per game (e.g. ['Role-playing', 'Action', 'Sci-fi']), so a single one-hot column per category would be wrong — a game can belong to several genres at once. MultiLabelBinarizer was used to create one binary column per genre (54 genre columns total).
- RUN:
from sklearn.preprocessing import MultiLabelBinarizer
import ast
mlb = MultiLabelBinarizer()
df['genres'] = df['genres'].apply(ast.literal_eval)
df_genres = pd.DataFrame(
 mlb.fit_transform(df['genres']),
 columns = mlb.classes_,
 index = df.index)

```
df = pd.concat([df.drop(columns=['genres']),df_genres],axis=1)
```

b. Platform support (worksOn)

- **worksOn is a dict of {Windows, Mac, Linux: bool}. This was unpacked into three separate binary columns.**

- RUN:

```
df['worksOn'] = df['worksOn'].apply(ast.literal_eval)
df['Windows'] = df['worksOn'].str['Windows']
df['Mac'] = df['worksOn'].str['Mac']
df['Linux'] = df['worksOn'].str['Linux']
df.drop('worksOn',axis=1,inplace=True)
```

c. Availability

- availability is a dict with isAvailable / isAvailableInAccount flags — unpacked the same way, then all boolean columns (isAvailable, isAvailableInAccount, Windows, Mac, Linux, isTBA, isInDevelopment, buyable, isComingSoon, isPriceVisible, isMovie, isGame, isWishlistable, isMod, isFree, isReviewable) were cast to int so the model can use them.

d. Frequency encoding for high-cardinality categoricals

- Developer and publisher have hundreds of unique text values, too many for one-hot encoding. Each was replaced with its own frequency count — i.e. how many games share that same developer/publisher/promo — which captures "is this a prolific studio" without exploding the column count.
- RUN:

```
df['developer'] = df['developer'].map(df['developer'].value_counts())
df['publisher'] = df['publisher'].map(df['publisher'].value_counts())
```

e. Sale status

- salesVisibility is a dict containing an isActive flag — extracted into a single salesActive binary column.
- RUN:

```
df['salesVisibility'] = df['salesVisibility'].apply(ast.literal_eval)
active = []
for i in df['salesVisibility']:
    active.append(i['isActive'])
df['salesActive'] = active
df.drop('salesVisibility',inplace=True,axis=1)
df['salesActive'] = df['salesActive'].astype(int)
```

f. **Category**

- Unlike genres, each game has exactly one category (Adventure, Action, Strategy, etc.), so standard one-hot encoding (with `drop_first=True` to avoid redundancy) was appropriate here.

Observations:

Now we are ready to train our ML model; No non-binary encoded columns, no null values and no duplicated rows.

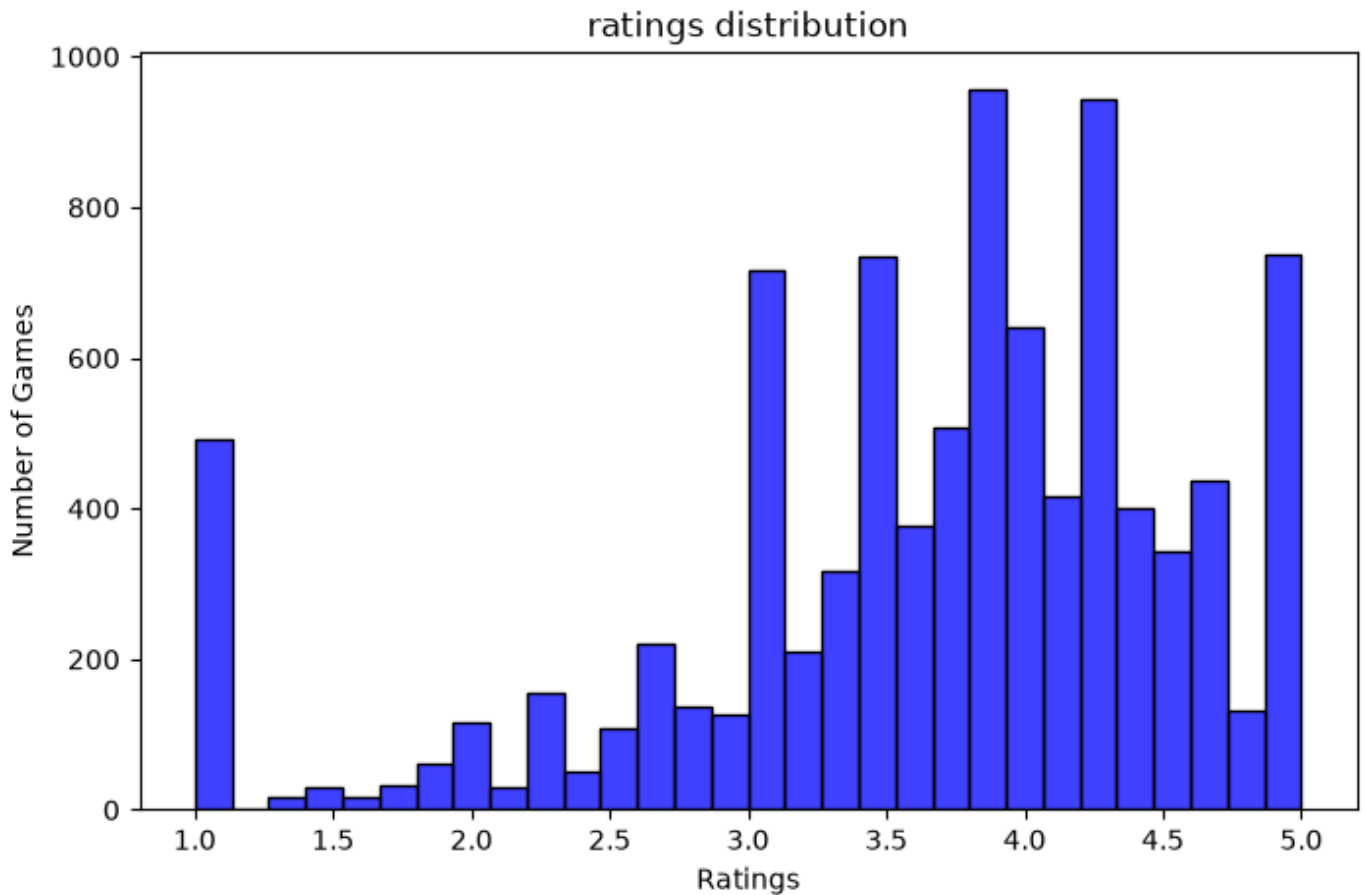
7. EDA(Exploratory Data Analysis):

Here, We need to see how some specific columns are related to game ratings using 4-5 graphs. We will do a comparison of the following:

a. Distribution of overall average ratings

RUN:

```
plt.figure(figsize=(8,5))
sns.histplot(df['overallAvgRating'],bins=30,color='Blue')
plt.xlabel("Ratings")
plt.ylabel("Number of Games")
plt.title("ratings distribution")
plt.show()
```



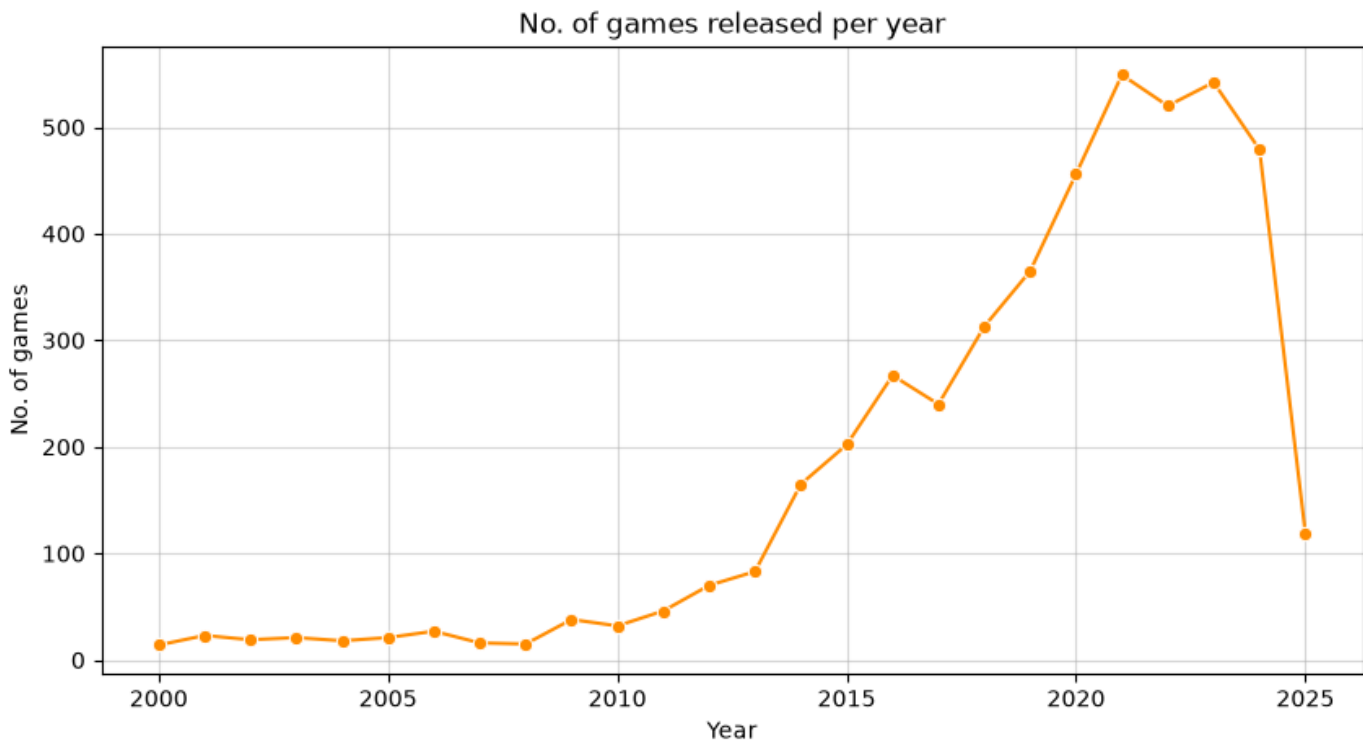
- Most games receive favorable ratings, indicating that the GOG catalog generally consists of well-reviewed titles.
- Very low-rated games are relatively uncommon.
- Users on the platform appear to rate games positively more often than negatively.

b. Games released per year

RUN:

```
release_years = pd.to_datetime(df_og['globalReleaseDate'],unit='s').dt.year
games_per_year = release_years.value_counts().sort_index()
games_per_year = games_per_year[games_per_year.index >= 2000]
plt.figure(figsize=(10,5))
sns.lineplot(x=games_per_year.index,y=games_per_year.values,marker='o',color='darkorange')
plt.xlabel("Year")
plt.ylabel("No. of games")
plt.title("No. of games released per year")
plt.grid(alpha=0.5)
```

plt.show()

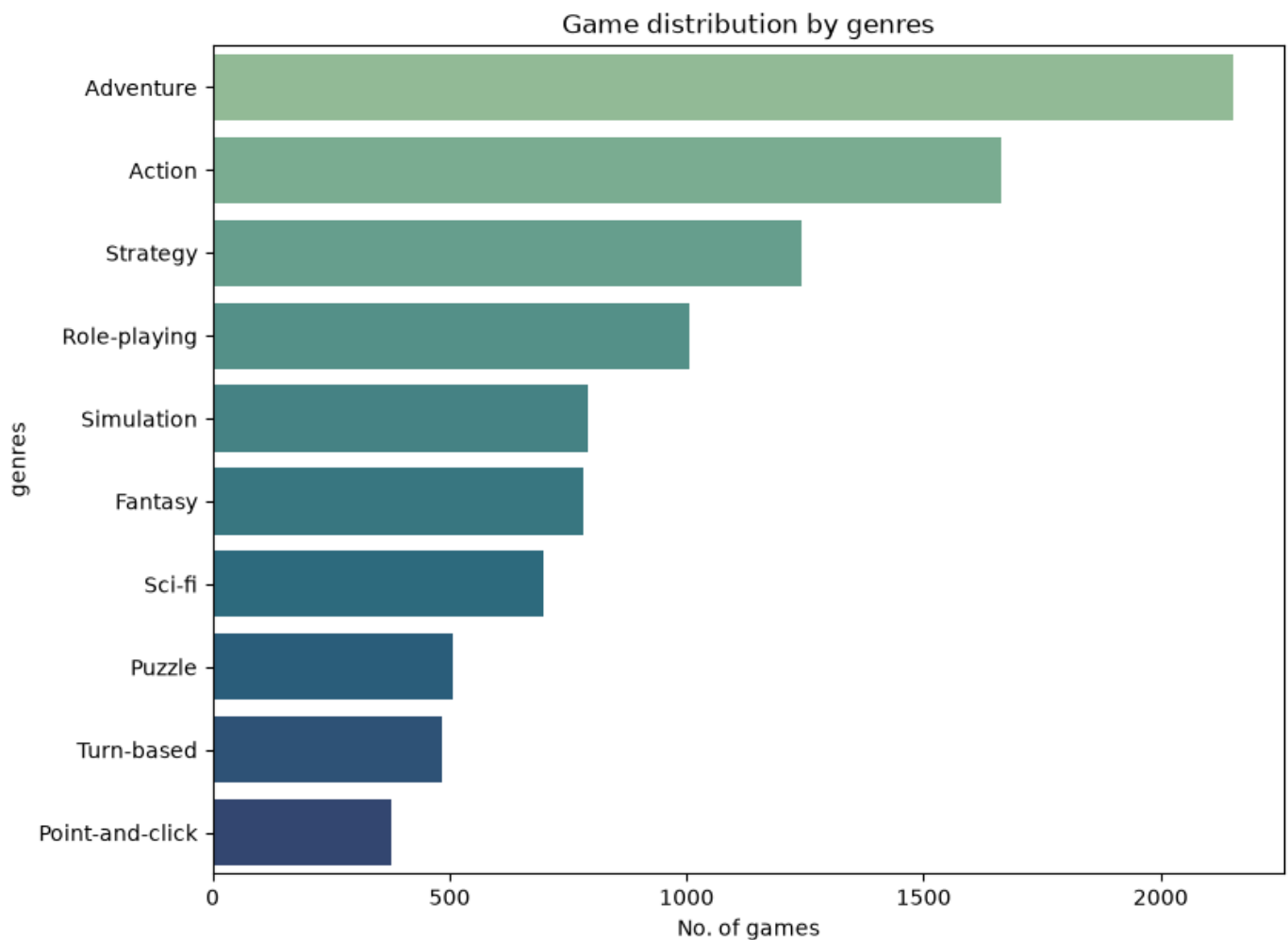


- The number of games released has generally increased over the years.
- Recent years contain a much larger selection of games than earlier years.
- The platform has expanded its game library significantly over time.

c. Top 10 Game Genres

RUN:

```
genre_series = df_og['genres'].apply(ast.literal_eval).explode()
top_genres = genre_series.value_counts().head(10)
plt.figure(figsize=(9,7))
sns.barplot(x=top_genres.values,y=top_genres.index,palette='crest')
plt.xlabel("Genres")
plt.xlabel("No. of games")
plt.title("Game distribution by genres")
plt.show()
```

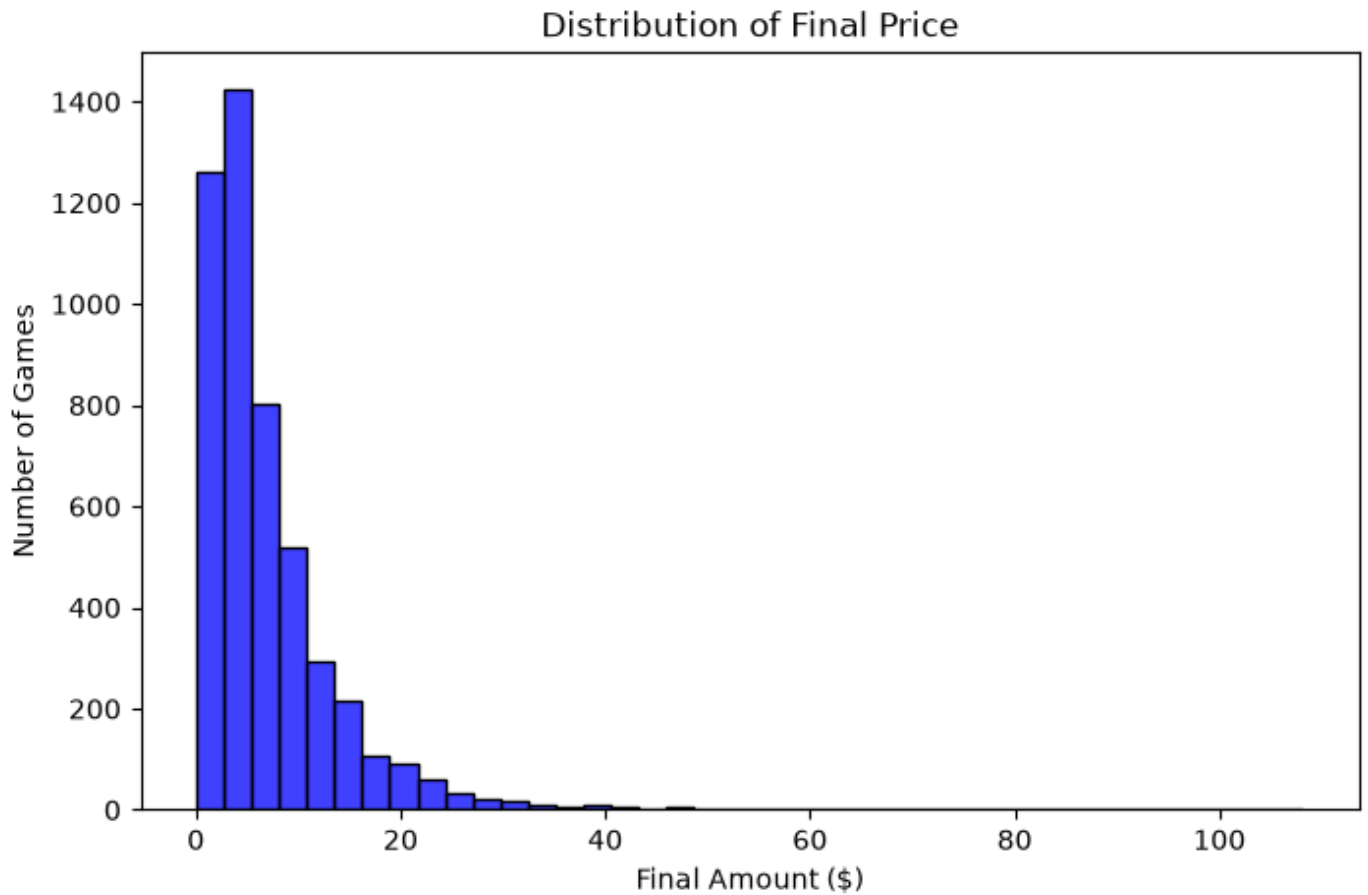



- A few genres dominate the platform's catalog.
- Action and Adventure-related games appear more frequently than niche genres.
- Players have a wide variety of games available, but certain genres are clearly more popular among developers.

d. Distribution of Final Price

RUN:

```
plt.figure(figsize=(8,5))
sns.histplot(df_og['finalAmount'], bins=40,color='Blue')
plt.title('Distribution of Final Price')
plt.xlabel('Final Amount ($)')
plt.ylabel('Number of Games')
```

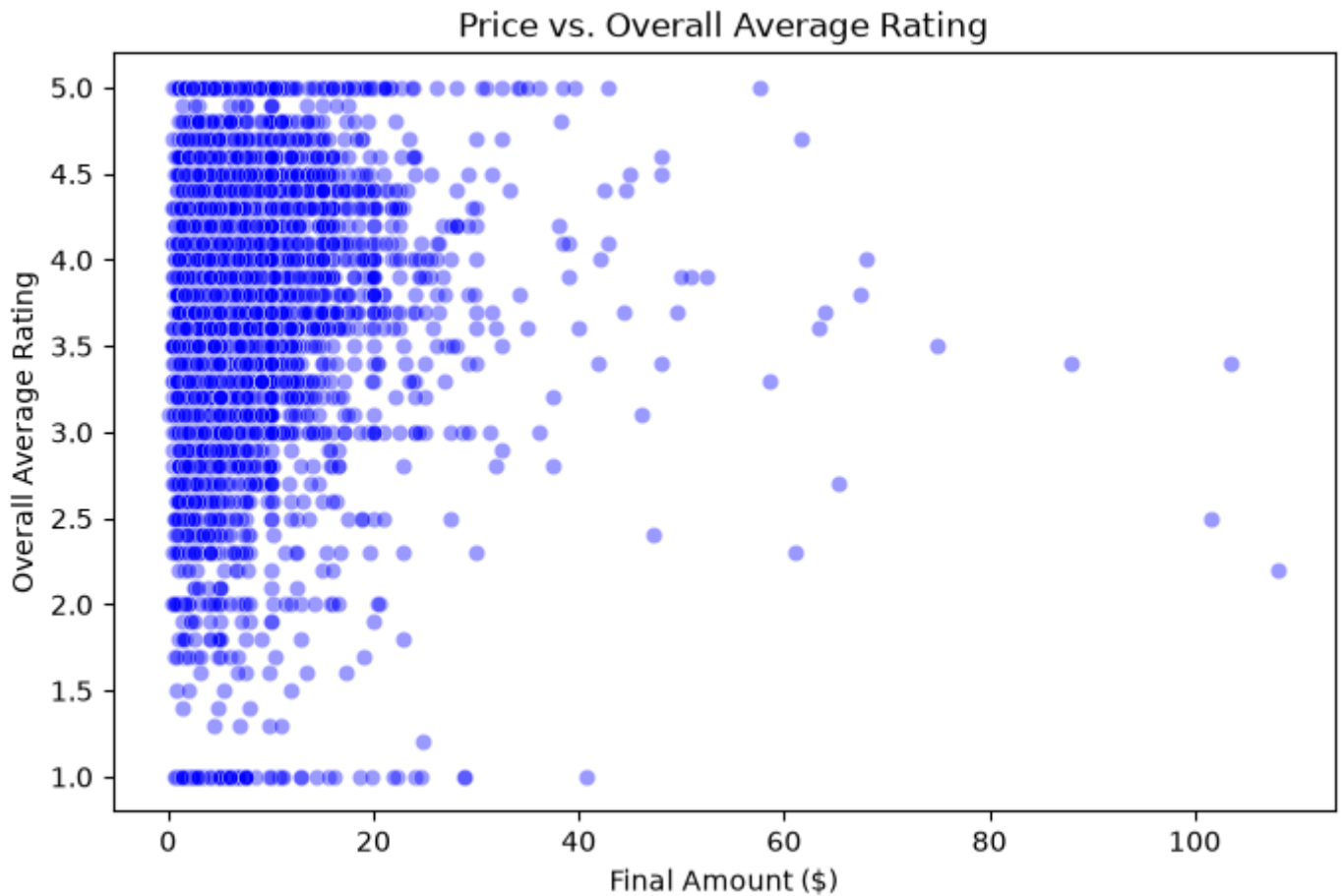


- Most games are priced in the lower price range.
- High-priced games are relatively uncommon.
- The platform offers a large number of affordable games, making it attractive to budget-conscious players.

e. Final Price vs Overall Average Rating

RUN:

```
filtered = df_og[df_og['overallAvgRating'] > 0]
plt.figure(figsize=(8,5))
sns.scatterplot(x=filtered['finalAmount'],
y=filtered['overallAvgRating'],alpha=0.4,color='Blue')
plt.title('Price vs. Overall Average Rating')
plt.xlabel('Final Amount ($)')
plt.ylabel('Overall Average Rating')
```

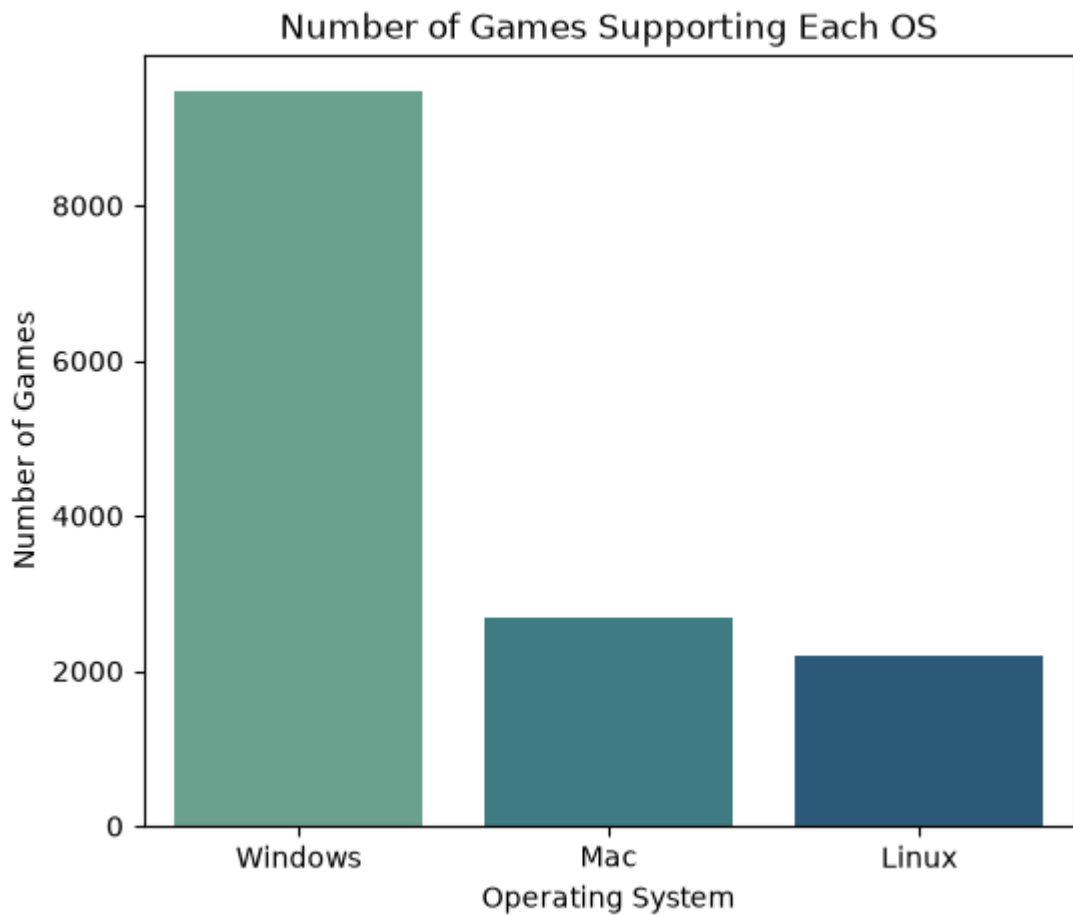


- Highly rated games can be found across different price ranges.
- Expensive games are not necessarily rated higher than cheaper games.
- Price alone does not appear to determine how positively a game is received.

f. Number of Games Supporting Each Operating System

RUN:

```
os_support = df[['Windows','Mac','Linux']].sum()
plt.figure(figsize=(6,5))
sns.barplot(x=os_support.index, y=os_support.values, palette='crest')
plt.title('Number of Games Supporting Each OS')
plt.xlabel('Operating System')
plt.ylabel('Number of Games')
```

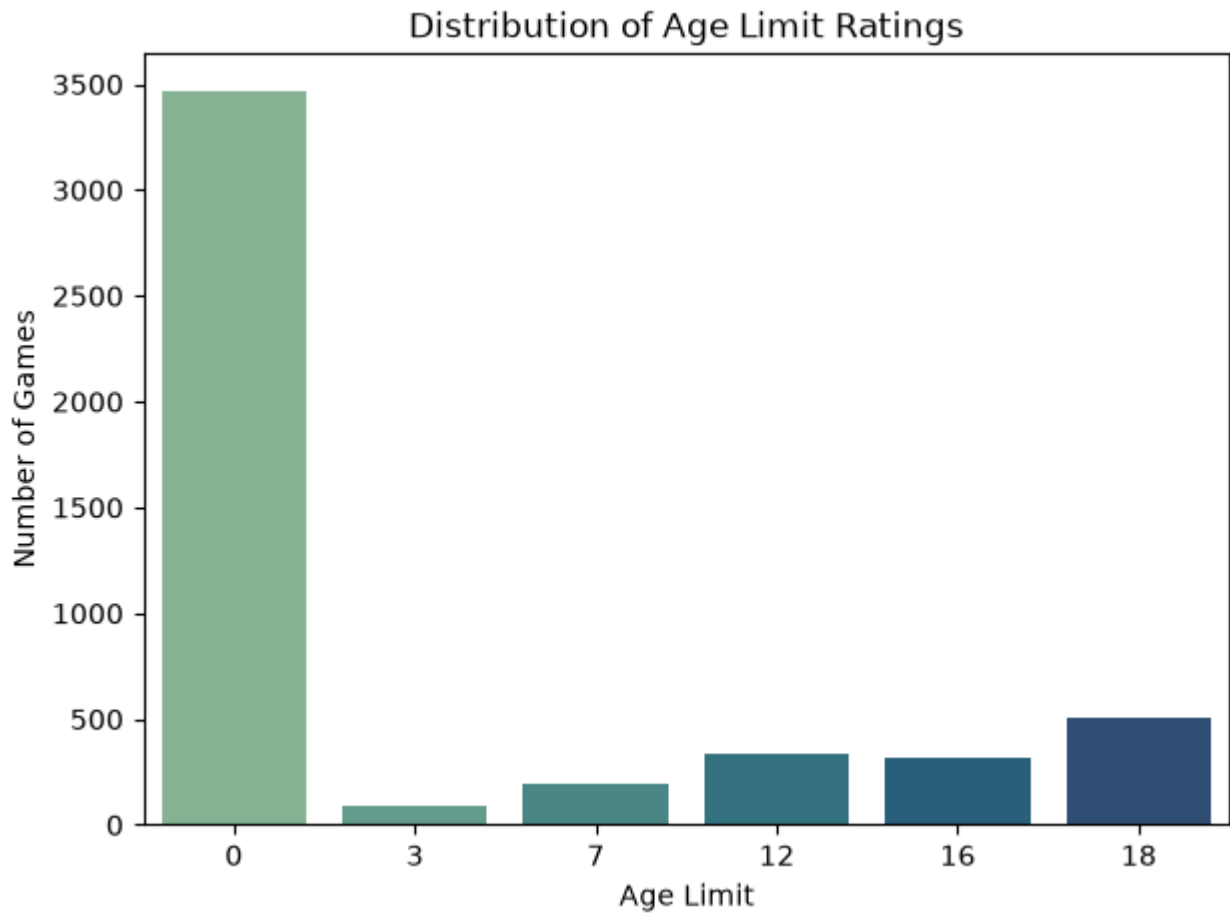


- Windows has the largest game support among all operating systems.
- macOS and Linux support is available for many games but is less common.
- Developers primarily target Windows while still offering compatibility for other platforms in selected titles.

g. Distribution of Age Limit Ratings

RUN:

```
plt.figure(figsize=(7,5))
sns.countplot(x=df_og['ageLimit'].sort_values(), palette='crest')
plt.title('Distribution of Age Limit Ratings')
plt.xlabel('Age Limit')
plt.ylabel('Number of Games')
plt.show()
```

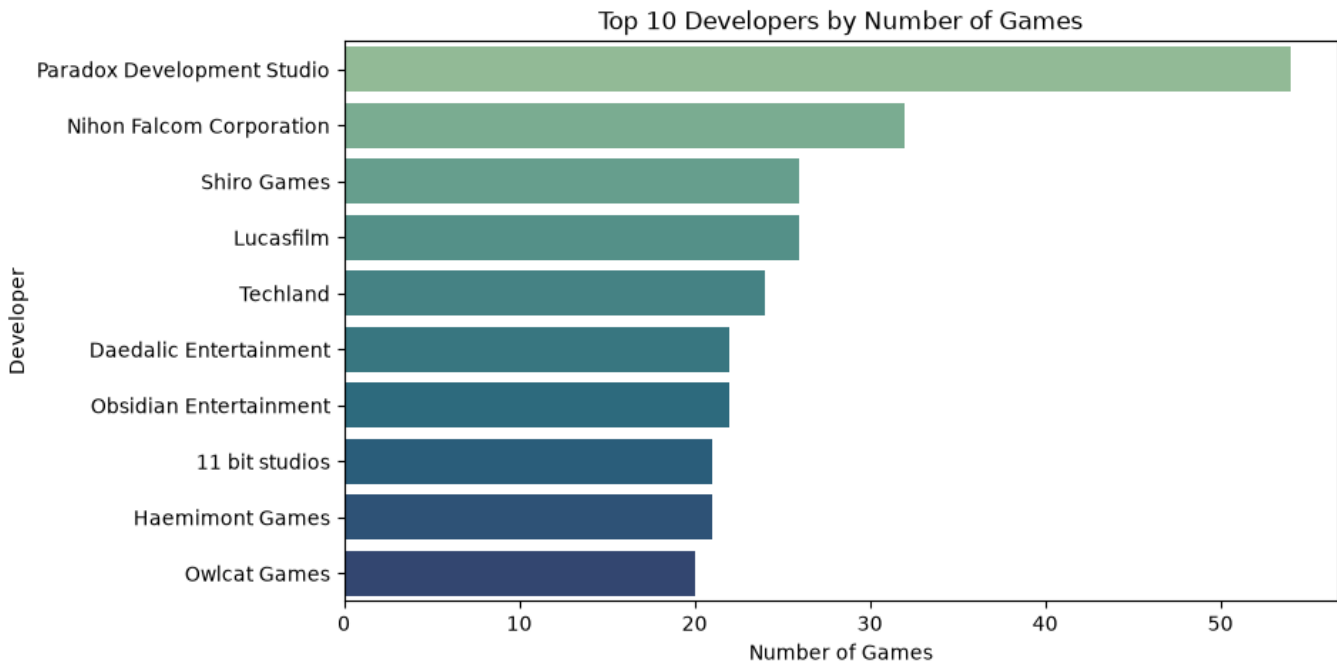


- Most games fall into a small number of age categories.
- Games suitable for teenagers and adults make up a significant portion of the catalogue.
- Very restrictive age ratings appear less frequently.

h. Top 10 Developers by Number of Games

RUN:

```
top_devs = df_og['developer'].value_counts().head(10)
plt.figure(figsize=(9,5))
sns.barplot(x=top_devs.values, y=top_devs.index, palette='crest')
plt.title('Top 10 Developers by Number of Games')
plt.xlabel('Number of Games')
plt.ylabel('Developer')
```



- A small group of developers contributes a large number of games to the platform.
- Most developers have only a few titles compared to the leading publishers.
- The catalog contains both established developers and many smaller studios.

8. Now we get into applying Random Forest algorithm onto this data:

We first split the data(80%trainin,20%testing)

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

9. Now we are going to apply the Random forest algorithm for this data. `x_train` will be X variable and `y_train` will be the y variable.

RUN:

```
from sklearn.ensemble import RandomForestClassifier
RF_model=RandomForestClassifier(n_estimators=500,stratify=y,
random_state=42)
RF.fit(x_train,y_train)
RF_pred = RF.predict(x_test)
print(classification_report(y_test,RF_pred))
print(confusion_matrix(y_test,RF_pred))
```

10. Now we need to find predictions based on test values of X:

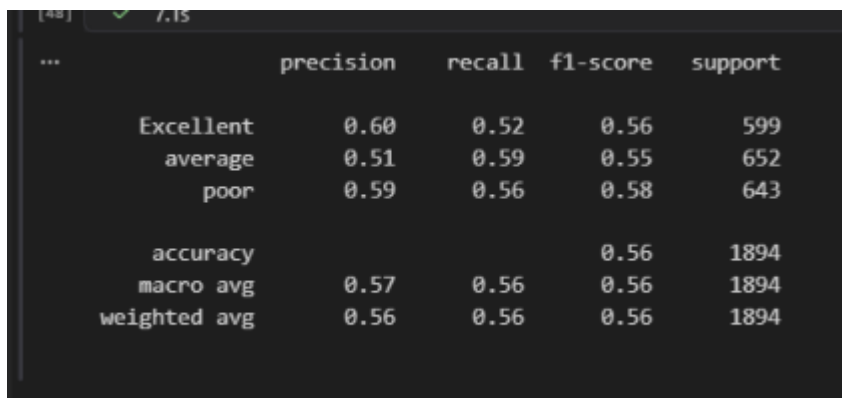
RUN:

```
Y_pred = RF_model.predict(x_test)
```

11. Then we need to check the accuracy of our model using metrics:

RUN:

```
from sklearn.metrics import classification_report
print(classification_report(y_test,pred))
```

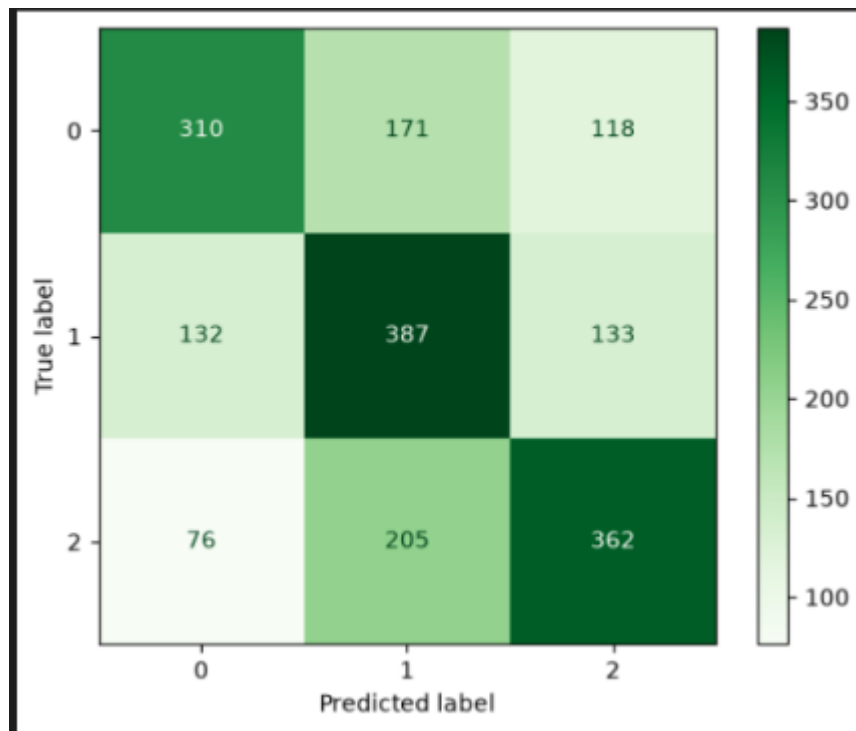


	precision	recall	f1-score	support
Excellent	0.60	0.52	0.56	599
average	0.51	0.59	0.55	652
poor	0.59	0.56	0.58	643
accuracy			0.56	1894
macro avg	0.57	0.56	0.56	1894
weighted avg	0.56	0.56	0.56	1894

12. Now we display the confusion matrix:

RUN:

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
cm = confusion_matrix(y_test, RF_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap=plt.cm.Greens)
plt.show()
```



Observations:

a. Poor:

- 310 were correctly predicted as poor
- 171 were incorrectly predicted as Average
- 118 were incorrectly predicted as Excellent

b. Average:

- 132 were incorrectly predicted as poor
- 387 were correctly predicted as Average
- 133 were incorrectly predicted as Excellent

c. Excellent:

- 76 were incorrectly predicted as poor
- 205 were incorrectly predicted as Average
- 362 were correctly predicted as Excellent